When Does Trade Protection Help? Evidence from the Rise of American Manufacturing

Vafa Behnam

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→ **This paper:** Revisit setting with new empirical design: find distance to technological frontier governs the sign of protection's effect

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  - Patent activity also shifts towards laggard products, suggesting reshaping of comp. advantage

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I also apply design to 2018 Trump tariffs episode, exploiting cross-state variation in pre-shock sector import volumes

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$\rightarrow$ Welfare effects of trade in GE ambiguous

Roadmap

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Data & Context

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Results & Interpretation

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- Local markets highly concentrated (Bateman et al. 1971); Median HHI of 0.56 in Hornbeck et al. 2026 data, more than double DoJ "highly concentrated" threshold of 0.25

Pricing Equation

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$$p_{iz} = p_z + t_z p_z + \tau_{iz}$$

p_z: pre-duty price at port; t_z: ad-valorem tariff; τ_{iz}: transport cost from port to county i

Specific Tariffs

Treatment Construction

Ad valorem equivalent of tariffs and transportation costs:

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- $\tau_{iz} > 0$: Import protection from transportation costs (e.g. Juhasz 2018, Pascali 2017)

Transportation Costs

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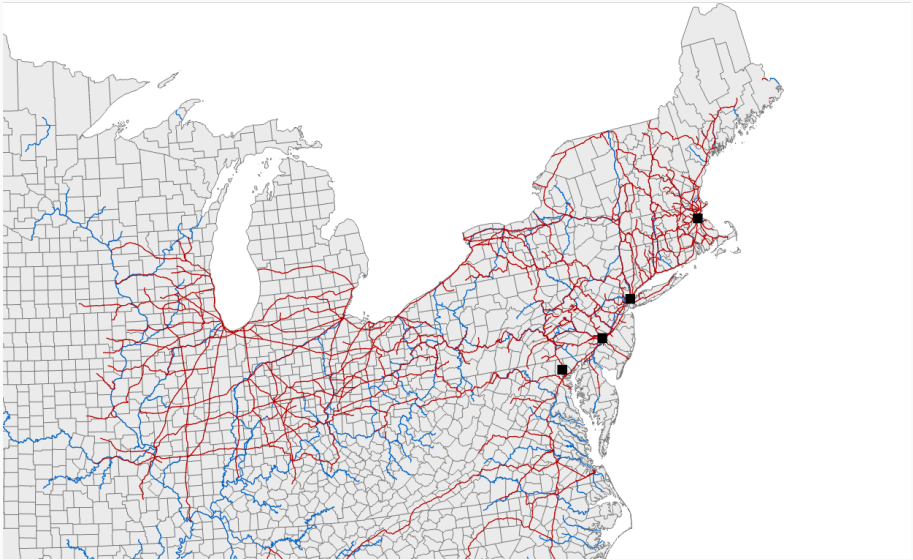
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Transportation Network 1860 (Atack 2015, 2016, 2017)



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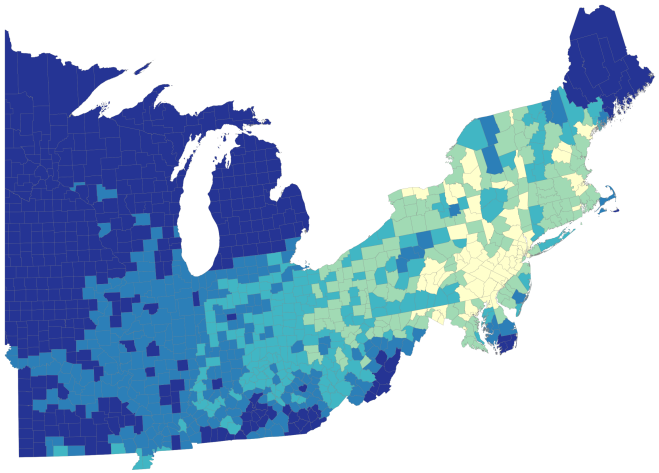
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  - Find $\hat{\omega} = 0.139$, examine robustness to other ωs

Nature of Transportation Costs

Costs Across Space: Pig Iron



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- I digitize tariffs t_z and port prices p_z from Foreign Commerce and Navigation of the US (1859, 1869)

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- Patent-based measure of technological gaps (similar to Akcigit, Ates and Impullitti 2024)

$$Dist_Front_z = \frac{\#\text{ British Patents}_{1855-1859,z}}{\#\text{ US Patents}_{1855-1859,z}}$$

  - British data from Walker Hanlon (2016), US data from Jim Shaw (accessed 2025)

Validation of Distance to Frontier

Cotton textiles: 1.31

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Glass: 3.39

- Scoville (1944): "Few innovations were of American origin. Technological development lagged appreciably behind that in Belgium, England, France, and Germany"

British Patent Data

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$$\Delta \log y_{iz} = \beta \Delta(\log(TP_{iz}) \times Dist_Front_z) + \gamma(\Delta \log(TP_{iz})) + \kappa X_{iz} + \Delta\delta_z + \Delta\delta_i + \Delta e_{iz}$$

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  - If Philly steel grows faster, infer output protection benefits steel growth

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- Idea: Isolate changes in protection coming from tariffs
- Then argue this component exogenous to local product trends

IV

To a first order approximation,

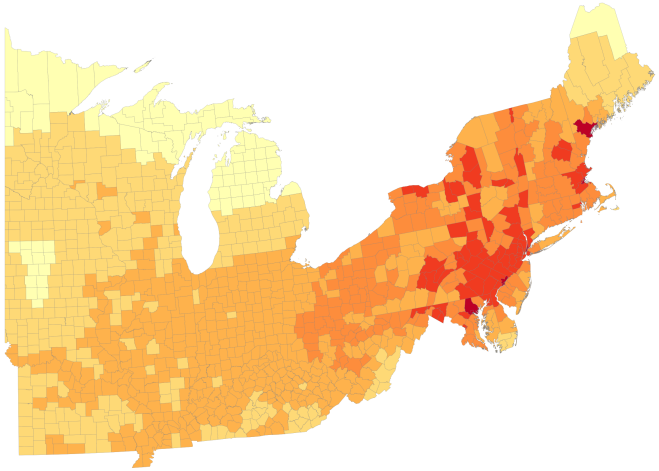
Derivation

$$\Delta \log(TP_{iz}) \approx \underbrace{STS^t_{iz} \Delta \log t_z}_{IV} + STS^\tau_{iz} \Delta \log \tau_{iz} - STS^\tau_{iz} \Delta \log p_z$$

where $STS^t_{iz} = \frac{t_z p_z}{t_z p_z + \tau_{iz}}$ is the share of tariff in total protection

Shift-Share vs. Shift-Exposure

IV Across Space



IV orthogonality condition:

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Interacted Term

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In words: *Nationwide* tariff changes not correlated with *local* product shocks, conditional on product and county FE

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In words: *Nationwide* tariff changes not correlated with *local* product shocks, conditional on product and county FE

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(Similar condition for shift-shares, Borusyak et al. 2022)

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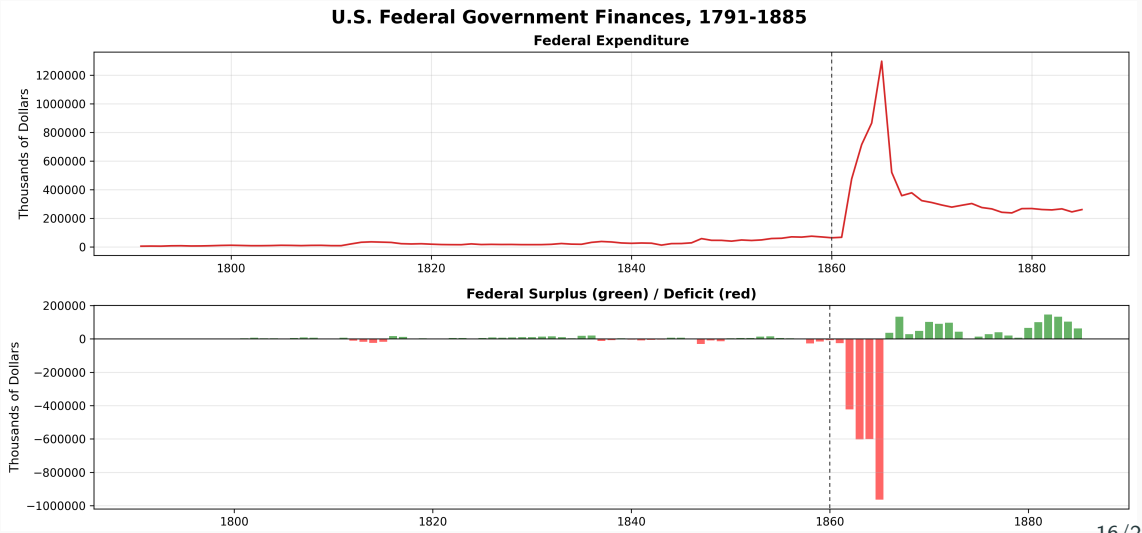
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Fiscal Environment

As % of GDP



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  - Allows me to isolate import protection from loss of export markets

Balance Tests

Balance Tests

	IV Output		IV Input	
Dependent Variable	Coef.	SE	Coef.	SE
Log Total Labor Initial	−1.921	(2.854)	−3.299	(2.051)
Log Establishments Initial	−2.015	(1.498)	−0.739	(1.098)
Log Capital Initial	−0.286	(2.369)	−2.545	(2.418)
Log Labor Productivity Initial	−0.694	(1.489)	0.539	(0.540)
Log Output Initial	−2.194	(1.908)	−3.190	(1.832)
Log Output per Estab Initial	−0.183	(1.929)	−2.405	(1.496)
Log Average Wage Initial	−0.635	(0.443)	0.242	(0.357)
Log Capital-Labor Ratio Initial	1.635	(2.696)	0.754	(1.094)

Notes: All specifications include icp-yearchange and product-yearchange fixed effects. Standard errors clustered at the icp and product level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

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	Δ Log Output	
	OLS	IV
	(1)	(2)
Δlog(TP) × Dist_Front	1.359***	1.493***
	(0.411)	(0.398)
Δlog(TP)	−5.144**	−4.424**
	(1.834)	(1.997)
Observations	833	833
Input TP Control	Yes	Yes
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→ Consistent with prior work on textiles (Irwin and Temin 2001), pig iron (Irwin 2000a) and glass (Taussig 1931, Klein and Meissner 2024)

Firm Size, Value Added and Factor Usage

	Δ Log Output/Estab.	Δ Log Value Added	Δ Log Labor	Δ Log Materials	Δ Log Capital
	(1)	(2)	(3)	(4)	(5)
Δlog(TP) × Dist_Front	1.818***	1.015**	1.383**	1.905***	−0.675
	(0.482)	(0.413)	(0.530)	(0.475)	(0.498)
Δlog(TP)	−6.049***	−3.588	−6.896**	−6.076**	−0.969
	(1.976)	(2.061)	(2.560)	(2.235)	(2.712)
Observations	833	832	833	833	833
Input TP Control	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes
First-stage F-stat (KP)	15.5	15.6	15.5	15.5	15.5

Notes: Standard errors two way clustered at the county and product level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Patent Analysis

	Δ Log Patents	
	OLS	IV
	(1)	(2)
Δlog(TP) × Distance to Frontier	1.175*	1.312**
	(0.596)	(0.572)
Δlog(TP)	−7.459*	−7.138*
	(3.872)	(3.537)
Observations	281	281
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→ Results suggest that pattern of comparative advantage itself changes

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  - Note no heterogeneity on cost side, e.g. advantage of backwardness (Gerschenkron 1962) $\frac{\partial \log \alpha_z}{\partial D_z} > 0$

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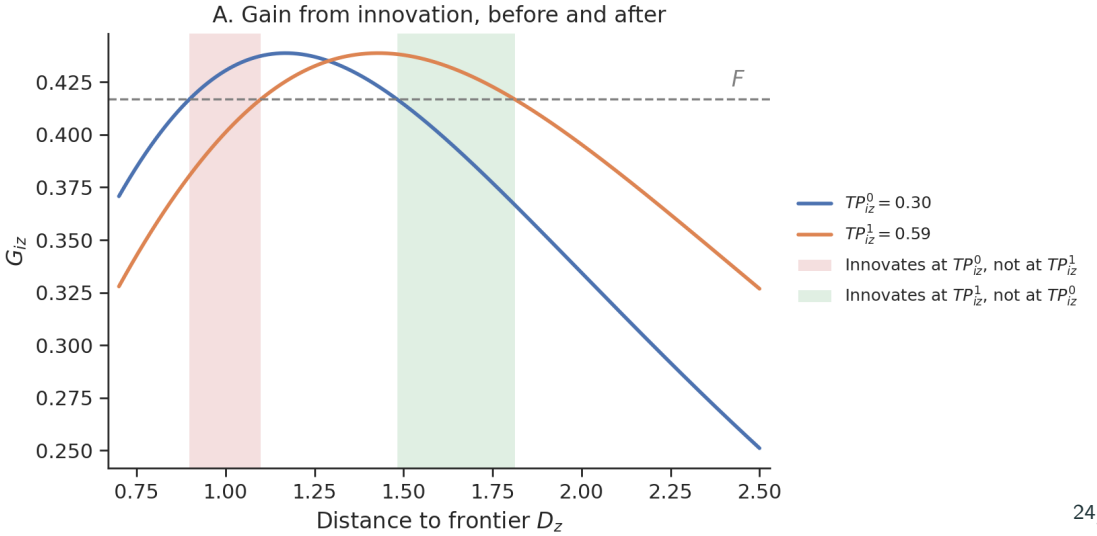
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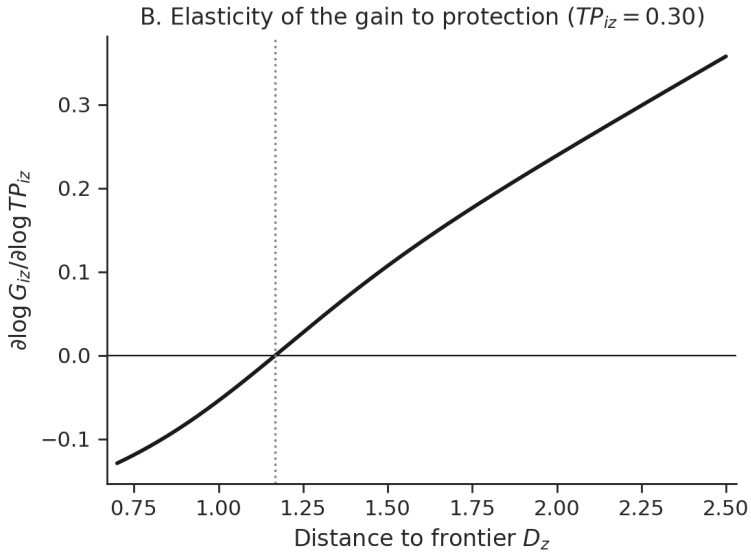
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- Markups can covary positively or negatively with innovation/upgrading

Level Gain From Upgrading Plot



Elasticity Plot



General Equilibrium

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- Market clearing:

$$Y_{iz} = E_{iz}\lambda_{iz} + E_{UK,z}\lambda_{UK,z} \quad \text{(goods)}$$

$$Y_i = w_i L_i + \Pi_i - F\sum_z I_{iz} \quad \text{(income)}$$

$$E_i = Y_i + TR_i \quad \text{(expenditure)}$$

where Π_i are profits, B_i is an exogenous deficit, and $TR_i = \rho_i TR$ is the tariff revenue rebated to the county by the Federal government

Single product formula. Let $W_i = \frac{E_i}{P_i}$ where E_i is total expenditures and P_i CES price index

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$$\Delta \log W_i = \underbrace{\frac{1}{1-\sigma}\Delta \log \lambda_i}_{\text{ACR term}} + \underbrace{\Delta I_i \log \alpha}_{\text{Upgrading term}} + \underbrace{\Delta \log(L_i - FI_i + \Pi_i^X)}_{\text{Resource cost}}.$$

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Roadmap

Introduction

Data & Context

Empirical Strategy & Identification

Results & Interpretation

Theory

Conclusion

Conclusion: Methodological Implications

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Other Interpretations

- Heterogeneous trade elasticities Details
- Pricing-to-market / exporter charging county-varying markups Details
- Quality upgrading via Alchian-Allen Effects Details
- Migration across counties Details
- Spatial demand spillovers Details

Welfare Effects of Trade: Multiproduct

$$\Delta \log W_i = \underbrace{\sum_z \chi_{iz} \tfrac{1}{1-\sigma} \Delta \log \lambda_{iz}}_{\text{ACR term}} + \underbrace{\sum_z \chi_{iz} \Delta I_{iz} \log \alpha}_{\text{upgrading effect}} + \underbrace{\Delta \log \left(L_i - F \sum_z I_{iz} + \sum_z \Pi_{iz}^X \right)}_{\text{resource cost}}$$

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→ Markup increases reduce welfare in products important for consumption, but raise it in products important for domestic profits

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$\rightarrow$ Net effect negative when $\frac{\Delta I_{iz}}{\Delta \log(TP_{iz})} < -\frac{1}{\log \alpha}$: innovation loss overwhelms substitution

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$\rightarrow$ $\hat{\beta}$ measures how this coastal-weighted average varies linearly with distance to the frontier

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- t_z^{SP}: Specific tariff

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$$AVE_{iz} := \frac{p_{iz} - p_z}{p_z} = t_z^{AV} + \frac{t_z^{SP}}{p_z} + \frac{\tau_{iz}}{p_z}$$

Treatment

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- Construct IO table using single product establishments from Atack et al. (2006)

Proof of Sufficiency of Identification Condition

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IV Orthogonality Condition

Orthogonality of Interacted Term

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$$\mathbb{E}(\Delta \log t_z \cdot Dist_Front_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z, STS_{iz}^t)$$

$$= Dist_Front_z \cdot \underbrace{\mathbb{E}(\Delta \log t_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z, STS_{iz}^t)}_{=0 \text{ by sufficient condition}} = 0$$

Orthogonality of Interacted Term

The sufficient condition also implies exogeneity of the interacted instrument:

$$\begin{aligned}&\mathbb{E}(\Delta \log t_z \cdot Dist_Front_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z, STS_{iz}^t) \\ &= Dist_Front_z \cdot \underbrace{\mathbb{E}(\Delta \log t_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z, STS_{iz}^t)}_{=0 \text{ by sufficient condition}} = 0\end{aligned}$$

Then, by iterated expectations,

$$\mathbb{E}(STS_{iz}^t \cdot \Delta \log t_z \cdot Dist_Front_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z) = 0$$

Shift-Share vs Shift-Exposure

Suppose we have some aggregate shifter Δg_z, e.g. tariffs or Chinese imports. s_{iz} is share of product z in county i. Shift-share:

$$\Delta \log y_i = \beta(\sum_z s_{iz} \times \Delta g_z) + \Delta e_i$$

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"Shift-exposure" (my paper):

$$\Delta \log y_{iz} = \beta(STS^t_{iz} \times \Delta g_z) + \Delta \delta_i + \Delta \delta_z + \Delta e_{iz}$$

Two key identification differences:

1. Can include product and county FE in shift exposure design
2. Error terms different (one is county-wide, other is county-product specific)

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Nature of Transportation Costs

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- Origin o, destination d, product z, time t

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- If $\omega = 0$, transportation costs are additive
  - Transportation costs purely a function of τ_{odt}, e.g. geographic distance and transportation infrastructure; same cost for all products

Transportation Costs

Nature of Transportation Costs II

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Transportation Cost Results

	Log Transport Cost
	(1)
Log Import Price ($\hat{\omega}$)	0.139^{***}
	(0.022)
Observations	124
Route-Year FE	Yes

Transportation Cost Results

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	(1)
Log Import Price ($\hat{\omega}$)	0.139***
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Route-Year FE	Yes

- Main results are robust to setting $\omega = 0$, $\omega = 0.0959$ (Lower bound of 95% CI) and $\omega = 0.182$ (Upper bound of 95% CI)

Transportation Costs

British Patent Data

Product	British Patents	Category
Cotton Textiles	464	First IR
Worsted Goods	464	First IR
Iron Bar	374	First/Second IR
Iron Pig	374	First/Second IR
Steel	374	Second IR
Paper	301	First IR
Paints	297	Second IR
Glove	222	Neither
Nails	214	First/Second IR
Boots	181	Neither
Hats	169	Neither
Beer	149	Neither
Whiskey	149	Neither
Rope	128	Neither
Glass	95	Neither

Dist to Frontier

Product	Distance to Frontier	British Patents	US Patents
Cotton Textiles	1.31	464	353
Worsted Goods	1.31	464	353
Boots	1.71	181	106
Hats	1.86	169	91
Iron Bar	2.53	374	148
Iron Pig	2.53	374	148
Steel	2.53	374	148
Paper	2.79	301	108
Beer	2.87	149	52
Whiskey	2.87	149	52
Glass	3.39	95	28
Gloves	3.64	222	61
Rope	3.88	128	33
Nails	4.98	214	43
Paints	7.82	297	38

Validation of Dist to Frontier

- Glass: 3.39
  - Scoville (1944): "The first glassmen in the United States naturally patterned their methods after those current in European countries, and they continued to do so throughout most of the nineteenth century. Few innovations were of American origin. Technological development lagged appreciably behind that in Belgium, England, France, and Germany"

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  - Jeremy (1981): By the early 19th century, this flow of technology was already moving in both directions across the Atlantic

Validation of Dist to Frontier

- Iron/steel 2.53:
  - Allen (1979): "In the middle of the nineteenth century Britain was the major supplier of iron and steel to the world market, while Germany and the United States were substantial importers... I shall demonstrate that Britain's mid-century export success was due to its superior technical efficiency"

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  - Temin (1964): "Before the Civil War, American manufacturers made only small quantities of steel. Because they were unable to master the demanding requirements to create steel through puddling, imports from England's Sheffield mills dominated the American market"

Validation of Dist to Frontier

- Paints: 7.82
  - Virtually all paint innovations in 19th century originated in Europe (Barnett 2004)
  - Sherwin-Williams becomes first US paint firm to employ a chemist in 1884. British paint firms had been founded by chemists decades earlier — Lewis Berger (1760), Winsor & Newton (1832)

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- Beer: 2.87
  - Pattinson (2014): "While at the beginning of the century, British breweries had led the world in technology and scale of operation, by the end of it the Americans were the ones at the bleeding edge of innovation."

Sample Share of Total Manufacturing

Table 1: Matched Sample Shares in 1860 Manufacturing

Measure	Share of Total Manufacturing
Output	27.93%
Workers	36.05%
Capital	32.42%
Value added	30.84%

Characteristics of Sample vs. Larger Sample

Table 2: Characteristics of the Sample Relative to Full Manufacturing Sample in 1860

Measure	All Manufacturing	Sample
Capital share	0.405	0.448
Labor share	0.147	0.160
Materials share	0.447	0.392
Output per establishment	14,427.48	21,560.35
Workers per establishment	9.66	18.52
Output per worker	1,493.37	1,164.28
Value added per establishment	6,256.82	10,319.28
Value added per worker	647.64	557.25
Capital per worker	766.22	694.16

Extensive Margin

$$1[y_{iz,1860} = 0,\ y_{iz,1870} > 0] = \beta[\Delta \log(TP_{iz}) \times Dist_Front_z] + \theta \Delta \log(TP_{iz}) + \gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz}$$

	$\mathbf{1}[y_{iz,1860} = 0,\ y_{iz,1870} > 0]$	
	OLS	IV
	(1)	(2)
Δlog(TP) $\times$ Dist_Front	-0.002	-0.007
	(0.013)	(0.022)
Δlog(TP)	0.033	0.082
	(0.067)	(0.126)
Observations	11,501	11,501
Input TP Control	Yes	Yes
County FE	Yes	Yes
Product FE	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	—	20.31

Robustness to Transportation Costs

	Δ Log Output		
	$\omega = 0$	$\omega = 0.096$	$\omega = 0.182$
	(1)	(2)	(3)
Δlog(TP) × Distance to Frontier	1.066***	1.340***	1.666***
	(0.288)	(0.354)	(0.452)
Δlog(TP)	-3.289*	-4.017**	-4.885**
	(1.557)	(1.820)	(2.225)
Observations	833	833	833
Input TP Control	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	10.6	14.0	16.9

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South Inclusive Growth Results

	Δ Log Output (1)
Δlog(TP) × Distance to Frontier	1.508***
	(0.406)
Δlog(TP)	-4.159*
	(2.109)
Observations	883
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-statistic (Kleibergen-Paap)	16.6

- Effect flips from negative to positive at:

$$Dist_Front_z = \frac{4.159}{1.508} \approx 2.76$$

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Excluding Union Slave States

	Δ Log Output
	(1)
Δlog(TP) × Distance to Frontier	1.203***
	(0.267)
Δlog(TP)	-3.634*
	(1.995)
Observations	771
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-statistic (Kleibergen-Paap)	19.2

- Effect flips from negative to positive at:

$$Dist_Front_z = \frac{3.63}{1.20} \approx 3.02$$

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Excluding Battle Counties

	Δ Log Output (1)
Δlog(TP) × Distance to Frontier	1.568***
	(0.406)
Δlog(TP)	-4.691**
	(2.037)
Observations	853
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-statistic (Kleibergen-Paap)	16.2

- Effect flips from negative to positive at:

$$Dist_Front_z = \frac{4.691}{1.568} \approx 2.99$$

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Excluding Cotton Textiles

	Δ Log Output (1)
Δlog(TP) × Distance to Frontier	1.663***
	(0.469)
Δlog(TP)	-4.833**
	(2.146)
Observations	743
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-statistic (Kleibergen-Paap)	15.9

- Effect flips from negative to positive at:

$$Dist_Front_z = \frac{4.83}{1.66} \approx 2.91$$

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Leave-One-Out Robustness

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Product Excluded	$\Delta \log(\text{TP}) \times Dist_Front$	$\Delta \log(\text{TP})$	KP F	N
Beer	1.823*** (0.468)	−5.839** (2.354)	12.09	614
Boots	1.640*** (0.451)	−4.589* (2.315)	16.00	592
Cotton Goods	1.659*** (0.468)	−4.821** (2.142)	15.82	743
Glass	1.502*** (0.428)	−4.342* (2.060)	21.42	797
Gloves	1.443*** (0.408)	−4.359* (2.035)	15.63	826
Hats	1.350*** (0.387)	−3.721* (2.052)	16.40	794
Iron Bar	1.348** (0.471)	−4.017* (2.092)	13.51	766
Iron Pig	1.174** (0.516)	−2.967** (1.321)	12.20	774
Nails	1.394*** (0.395)	−4.511** (1.870)	10.17	808
Paints	1.938** (0.881)	−5.751* (2.833)	13.83	823
Paper	1.528*** (0.421)	−4.287** (2.120)	15.55	716
Rope	1.113*** (0.289)	−3.066* (1.525)	15.44	798
Steel	1.503*** (0.403)	−4.465** (2.012)	15.44	830
Whiskey	1.674*** (0.503)	−5.935** (2.355)	15.65	729
Worsted Goods	1.498*** (0.400)	−4.430** (2.004)	15.45	830

SEs in parentheses. Each row excludes one product. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Dropping Top 5 Cities

	Δ Log Output	
	Top 5 Cities Excluded	Top 5 Cities + Neighbors Excluded
	(1)	(2)
Δlog(TP) × Distance to Frontier	1.690***	1.527***
	(0.544)	(0.485)
Δlog(TP)	-4.395*	-4.250**
	(2.169)	(1.966)
Observations	801	739
Input TP Control	Yes	Yes
County FE	Yes	Yes
Product FE	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	15.7	18.1

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Excluding Counties with Military Purchases

	Δ Log Output (1)
Δlog(TP) × Distance to Frontier	1.695***
	(0.549)
Δlog(TP)	-5.073**
	(2.339)
Observations	800
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-statistic (Kleibergen-Paap)	13.6

- Effect flips from negative to positive at:

$$Dist_Front_z = \frac{5.07}{1.69} \approx 2.99$$

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Excluding Counties with Military Purchases + Neighbors

	Δ Log Output (1)
Δlog(TP) × Distance to Frontier	1.434***
	(0.425)
Δlog(TP)	-4.383*
	(2.087)
Observations	765
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-statistic (Kleibergen-Paap)	14.2

- Effect flips from negative to positive at:

$$Dist_Front_z = \frac{4.38}{1.43} \approx 3.06$$

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Log Distance to Frontier

	Δ Log Output
	(1)
Δlog(TP) $\times$ log(Dist_Front)	6.156**
	(2.331)
Δlog(TP)	-6.455*
	(3.055)
Observations	833
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-statistic (Kleibergen-Paap)	18.0

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Dropping Top 5 Cities

Back

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Δlog(TP)	-4.395*	-4.250**
	(2.169)	(1.966)
Observations	801	739
Input TP Control	Yes	Yes
County FE	Yes	Yes
Product FE	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	15.7	18.1

Regressions with Initial Variables

Back

	Δ Log Output			
	(1)	(2)	(3)	(4)
	Output Init.	Estab. Init.	Labor Init.	Cap./Labor Init.
Δlog(TP) × Distance to Frontier	1.474***	1.707***	1.594***	1.977***
	(0.418)	(0.446)	(0.441)	(0.529)
Δlog(TP)	−5.231**	−5.962***	−5.499**	−4.993*
	(1.821)	(1.920)	(1.869)	(2.382)
Observations	833	833	833	833
Input TP Control	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	15.57	15.12	15.72	16.23

Growth Results

	Δ Log Output			
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
Δlog(TP) × Dist_Front			1.359***	1.493***
			(0.411)	(0.398)
Δlog(TP)	−0.770	−0.281	−5.144**	−4.424**
	(1.104)	(1.194)	(1.834)	(1.997)
Observations	833	833	833	833
Input TP Control	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes
First-stage F-stat (Kleibergen-Paap)	—	16.5	—	15.5

Effect flips from negative to positive at: $Dist_Front_z = \frac{4.434}{1.496} \approx 2.96$

Why Didn't Major Trade Partners Retaliate?

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  - Retaliatory measures debated — especially during the Civil War — but never passed

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  - Germany had yet to unify in this period (1871), inhibiting unified responses
  - The two major German ports, Hamburg and Bremen, were not part of the Zollverein customs union
  - In addition, every member in the Zollverein had a veto until 1867
- **France**

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- **France**
  - Seemed more concerned with trade with other European nations and colonial empire, not US

Historical Discussion on Spatial Impact of Tariffs

"If the duty on pig iron were entirely removed, the American producer in the interior would still enjoy a protection in the cost of transportation . . . that intervene between the place of production and the port of entry . . . which circumstances renders the transport of a single pound of foreign pig iron to any considerable distance into the interior a matter of ordinary commercial impossibility"

— Commissioner David A. Wells (1869)

Historical Discussion on Spatial Impact of Tariffs

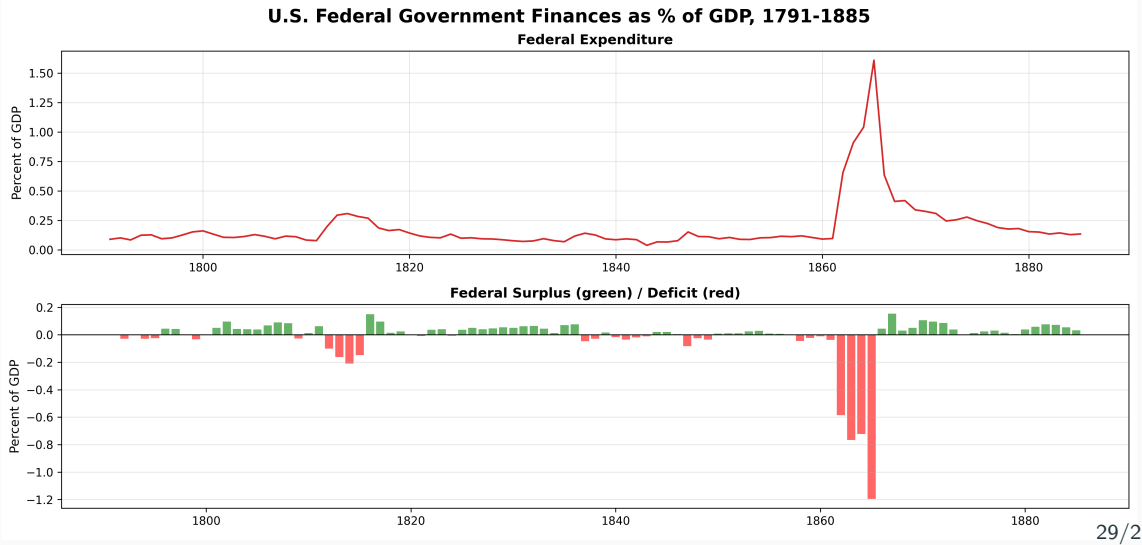
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— Commissioner David A. Wells (1869)

"The cost of transportation from the sea-board to the interior is such that, even in the absence of the duty, steel rails would be imported only to supply railways near tide-water"

— Frank Taussig (1915)

Back to Contribution


U.S. Federal Government Finances as % of GDP, 1791-1885
Federal Expenditure
Percent of GDP
1.50
1.25
1.00
0.75
0.50
0.25
0.00
1800
1820
1840
1860
1880
Federal Surplus (green) / Deficit (red)
Percent of GDP
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
1800
1820
1840
1860
1880

Controlling for Trade Elasticities

	Δ Log Output			
	(1)	(2)	(3)	(4)
	Median	CP (2015)	Shapiro (2016)	GYY (2018)
$\Delta\log(\text{TP}) \times$ Distance to Frontier	2.650**	1.585***	3.586**	1.995***
	(1.027)	(0.433)	(1.370)	(0.695)
$\theta_z \times \Delta\log(\text{TP})$	0.851	1.426	0.993*	0.395
	(0.602)	(2.030)	(0.539)	(0.491)
$\Delta\log(\text{TP})$	−15.009*	−9.242	−23.539*	−9.253
	(7.914)	(6.883)	(11.410)	(6.052)
Observations	833	833	833	833
Input TP Control	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	7.59	3.55	25.62	7.71

Alternative Interpretations

First-Order Approximation of $\Delta \log(TP_{iz})$ (cont.)

$$\frac{1}{TP_{iz}}\, d(t_z) = \frac{t_z}{t_z + \frac{\tau_{iz}}{p_z}}\, d\log t_z = \underbrace{\frac{p_z\, t_z}{p_z\, t_z + \tau_{iz}}}_{STS^t_{iz}}\, d\log t_z$$

$$\frac{1}{TP_{iz}}\, \frac{\tau_{iz}}{p_z}\, (d\log \tau_{iz} - d\log p_z) = \underbrace{\frac{\tau_{iz}}{t_z\, p_z + \tau_{iz}}}_{STS^\tau_{iz}}\, (d\log \tau_{iz} - d\log p_z)$$

$$\boxed{d\log(TP_{iz}) = STS^t_{iz}\; d\log t_z + STS^\tau_{iz}\; (d\log \tau_{iz} - d\log p_z)}$$

Back to IV

(Im)balance Test Without Product-Time FE

Back

	IV Output		IV Input	
Dependent Variable	Coef.	SE	Coef.	SE
Log Total Labor (first)	−0.143	(0.151)	−0.885*	(0.444)
Log Establishments (first)	0.018	(0.056)	−0.232	(0.170)
Log Capital (first)	0.032	(0.128)	−0.877*	(0.437)
Log Labor Productivity (first)	0.264***	(0.089)	0.331	(0.197)
Log Output (first)	0.208*	(0.107)	−0.430	(0.355)
Log Output per Estab (first)	0.189	(0.131)	−0.192	(0.397)
Log Average Wage (first)	0.062**	(0.024)	0.192**	(0.081)
Log Capital-Labor Ratio (first)	0.175	(0.124)	0.008	(0.272)

Notes: All specifications include icp-yearchange fixed effects only. Standard errors clustered at the icp and product level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

(Im)balance Test Without County-Time FE

Back

	IV Output		IV Input	
Dependent Variable	Coef.	SE	Coef.	SE
Log Total Labor (first)	11.473**	(5.105)	1.463	(2.350)
Log Establishments (first)	5.759*	(2.803)	1.758	(1.095)
Log Capital (first)	15.028*	(7.885)	2.201	(2.839)
Log Labor Productivity (first)	1.556	(1.683)	1.071**	(0.462)
Log Output (first)	14.926*	(7.171)	2.071	(2.819)
Log Output per Estab (first)	9.155*	(4.508)	0.370	(2.046)
Log Average Wage (first)	−0.060	(0.607)	0.175	(0.159)
Log Capital-Labor Ratio (first)	3.555	(3.071)	0.738	(0.793)

Notes: All specifications include product-yearchange fixed effects only. Standard errors clustered at the icp and product level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Spatial Impact of Tariffs – Theory

CES demand for domestic varieties in county i, product z at time s:

$$X_{iizs}^{d} = p_{iizs}^{1-\sigma} P_{izs}^{\sigma-1} E_{izs}$$

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where $\lambda_{ifzs} = \frac{p^{1-\sigma}_{ifzs}}{P^{1-\sigma}_{izs}}$ is the import expenditure share

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- $\frac{p_{fzs} t_{zs}}{p_{fzs}(1+t_{zs})+\tau_{iz}}$ decreasing in τ_{iz}
- λ_{ifzs} also tends to be higher for low τ_{iz}

$\rightarrow$ Coastal counties (low τ_{iz}) respond more to a national tariff increase

Spatial Impact of Tariffs – AIDS Demand

AIDS expenditure share on domestic goods (Fajgelbaum and Khandelwal 2016):

$$s_{izs}^{d} = \alpha_z + \gamma_z \ln\left(\frac{p_{izs}^{f}}{p_{izs}^{d}}\right) + \beta_z \ln\left(\frac{E_{izs}}{a(p)}\right)$$

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$\rightarrow$ Same spatial gradient: coastal counties receive larger demand shocks

$\rightarrow$ Non-homotheticities amplify the effect relative to CES

Numerical Example: Protection Reduces Output

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Set $\sigma = 8$, $E_{iz} = 1$, $\alpha = 3$, $D_z = 1.05$, $F = 0.4343$, with TP_{iz} rising from 0.25 to 0.35

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$\rightarrow$ Output falls 28.5% despite higher protection: the lost innovation more than offsets the direct protective effect

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→ Results also robust to directly controlling for $\theta_z \times \Delta \log(TP_{iz})$

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   - My framework and most others predict positive passthrough, whether complete or incomplete
2. No innovation channel means no prediction for patenting
   - Yet patents shift with the same frontier gradient as output

Alchian-Allen Effects

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3. No reason for either effect to differ by distance to frontier

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$\rightarrow$ $\hat{\theta}$ biased upward, so negative estimates are conservative

Migration Across Counties

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Welfare with Tariff Revenue

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Revenue collected nationally, rebated to county i in share ρ_i:

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The exact welfare formula becomes

$$\Delta \log W_i = \frac{1}{1-\sigma}\Delta \log \lambda_i + \Delta I_i \log \alpha + \Delta \log(L_i - FI_i + \Pi_i^X) + \underbrace{\Delta \log\left(1 + \frac{\rho_i TR}{L_i - FI_i + \Pi_i^X}\right)}_{\text{tariff-revenue effect}}.$$

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When national revenue is positive in both equilibria,

$$\Delta \log TR = \log\left(\frac{\sum_j \sum_z \frac{t'_z}{1+t'_z}(1-\lambda'_{jz})E'_{jz}}{\sum_j \sum_z \frac{t_z}{1+t_z}(1-\lambda_{jz})E_{jz}}\right),$$

where primes denote the post-liberalization equilibrium.

Numerical Example: Welfare Loss from Trade

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$\rightarrow$ Liberalization lowers welfare by 0.542% by shutting down innovation.

Holding innovation fixed, the same liberalization raises welfare by 10.51%.